

CS229
Problem Set 4

Connor Yan

July 31, 2026

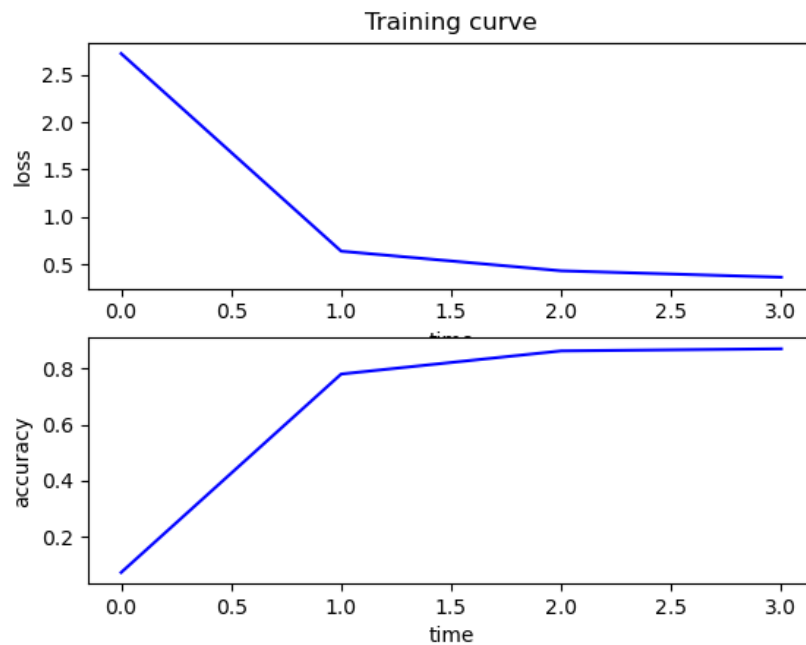


Figure 1: Training curve of loss and accuracy

Problem 1

(a) See the code.

(a) See the code and Figure 1.

Problem 2

(a)

$$\begin{aligned} \mathbb{E}_{\substack{s \sim p(s) \\ a \sim \pi_0(s,a)}} \left[\frac{\pi_1(s,a)}{\pi_0(s,a)} R(s,a) \right] &= \sum_{(s,a)} \frac{\pi_1(s,a)}{\pi_0(s,a)} R(s,a) p(s) \pi_0(s,a) \\ &= \sum_{(s,a)} \pi_1(s,a) R(s,a) p(s) \\ &= \mathbb{E}_{\substack{s \sim p(s) \\ a \sim \pi_1(s,a)}} [R(s,a)] \end{aligned}$$

(b)

$$\begin{aligned} \frac{\mathbb{E}_{\substack{s \sim p(s) \\ a \sim \pi_0(s,a)}} \left[\frac{\pi_1(s,a)}{\pi_0(s,a)} R(s,a) \right]}{\mathbb{E}_{\substack{s \sim p(s) \\ a \sim \pi_0(s,a)}} \left[\frac{\pi_1(s,a)}{\pi_0(s,a)} \right]} &= \frac{\sum_{(s,a)} \frac{\pi_1(s,a)}{\pi_0(s,a)} R(s,a) p(s) \pi_0(s,a)}{\sum_{(s,a)} \frac{\pi_1(s,a)}{\pi_0(s,a)} p(s) \pi_0(s,a)} \\ &= \frac{\sum_{(s,a)} R(s,a) p(s) \pi_1(s,a)}{\sum_{(s,a)} p(s) \pi_1(s,a)} \\ &= \mathbb{E}_{\substack{s \sim p(s) \\ a \sim \pi_1(s,a)}} [R(s,a)] \end{aligned}$$

(c) Suppose we have n samples. Then the weighted importance sampling estimator has the form

$$\frac{\sum_{i=1}^n \frac{\pi_1(s_i, a_i)}{\pi_0(s_i, a_i)} R(s_i, a_i)}{\sum_{i=1}^n \frac{\pi_1(s_i, a_i)}{\pi_0(s_i, a_i)}}$$

Consider the case that there is only one sample, i.e, $n = 1$.

$$\begin{aligned} \frac{\sum_{i=1}^n \frac{\pi_1(s_i, a_i)}{\pi_0(s_i, a_i)} R(s_i, a_i)}{\sum_{i=1}^n \frac{\pi_1(s_i, a_i)}{\pi_0(s_i, a_i)}} &= \frac{\frac{\pi_1(s_1, a_1)}{\pi_0(s_1, a_1)} R(s_1, a_1)}{\frac{\pi_1(s_1, a_1)}{\pi_0(s_1, a_1)}} \\ &= R(s_1, a_1) \end{aligned}$$

The expectation is

$$\mathbb{E}_{\substack{s_1 \sim p(s_1) \\ a_1 \sim \pi_0(s_1, a_1)}} [R(s_1, a_1)] = \mathbb{E}_{\substack{s \sim p(s) \\ a \sim \pi_0(s, a)}} [R(s, a)]$$

Which is not quite likely to be equal to the objective value $\mathbb{E}_{\substack{s \sim p(s) \\ a \sim \pi_1(s, a)}} [R(s, a)]$.

(d)

i.

$$\begin{aligned} &\mathbb{E}_{\substack{s \sim p(s) \\ a \sim \pi_0(s, a)}} \left[\mathbb{E}_{a \sim \pi_1(s, a)} [\hat{R}(s, a)] + \frac{\pi_1(s, a)}{\pi_0(s, a)} (R(s, a) - \hat{R}(s, a)) \right] \\ &= \sum_{(s, a)} \left[\frac{\pi_1(s, a)}{\pi_0(s, a)} (R(s, a) - \hat{R}(s, a)) + \sum_a \hat{R}(s, a) \pi_1(s, a) \right] p(s) \pi_0(s, a) \\ &= \sum_{(s, a)} p(s) \pi_1(s, a) (R(s, a) - \hat{R}(s, a)) + \sum_{(s, a)} p(s) \pi_0(s, a) \sum_a \hat{R}(s, a) \pi_1(s, a) \\ &= \sum_{(s, a)} p(s) \pi_1(s, a) (R(s, a) - \hat{R}(s, a)) + \sum_s p(s) \sum_a \hat{R}(s, a) \pi_1(s, a) \\ &= \sum_{(s, a)} p(s) \pi_1(s, a) [R(s, a) - \hat{R}(s, a) + \hat{R}(s, a)] \\ &= \mathbb{E}_{\substack{s \sim p(s) \\ a \sim \pi_1(s, a)}} [R(s, a)] \end{aligned}$$

ii.

$$\begin{aligned} \mathbb{E}_{\substack{s \sim p(s) \\ a \sim \pi_0(s, a)}} \left[\mathbb{E}_{a \sim \pi_1(s, a)} [R(s, a)] + \frac{\pi_1(s, a)}{\hat{\pi}_0(s, a)} (R(s, a) - R(s, a)) \right] &= \mathbb{E}_{\substack{s \sim p(s) \\ a \sim \pi_0(s, a)}} [\mathbb{E}_{a \sim \pi_1(s, a)} [R(s, a)]] \\ &= \sum_{(s, a)} p(s) \pi_0(s, a) \sum_a R(s, a) \pi_1(s, a) \\ &= \sum_s p(s) \sum_a R(s, a) \pi_1(s, a) \\ &= \mathbb{E}_{\substack{s \sim p(s) \\ a \sim \pi_1(s, a)}} [R(s, a)] \end{aligned}$$

(e)

i. Importance sampling estimator since π is easier to be modelled.

ii. Regression estimator since R is easier to be modelled.

Problem 3

Note that

$$f_u(x) = u \cdot u^T x$$

Hence,

$$\begin{aligned}
 \arg \min_{u: u^T u = 1} \sum_{i=1}^m \|x^{(i)} - f_u(x^{(i)})\|_2^2 &= \arg \min_{u: u^T u = 1} \sum_{i=1}^m \|x^{(i)} - u \cdot u^T x^{(i)}\|_2^2 \\
 &= \arg \min_{u: u^T u = 1} \sum_{i=1}^m \|(I - uu^T)x^{(i)}\|_2^2 \\
 &= \arg \min_{u: u^T u = 1} \sum_{i=1}^m \left((I - uu^T)x^{(i)} \right)^T (I - uu^T)x^{(i)} \\
 &= \arg \min_{u: u^T u = 1} \sum_{i=1}^m (x^{(i)})^T (I - uu^T)^T (I - uu^T)x^{(i)} \\
 &= \arg \min_{u: u^T u = 1} \sum_{i=1}^m (x^{(i)})^T (I - 2uu^T + uu^T uu^T)x^{(i)} \\
 &= \arg \min_{u: u^T u = 1} \sum_{i=1}^m (x^{(i)})^T (u^T u - 2)uu^T x^{(i)} \\
 &= \arg \min_{u: u^T u = 1} \sum_{i=1}^m (x^{(i)})^T (1 - 2)uu^T x^{(i)} \\
 &= \arg \max_{u: u^T u = 1} \sum_{i=1}^m (x^{(i)})^T uu^T x^{(i)} \\
 &= \arg \max_{u: u^T u = 1} \sum_{i=1}^m ((x^{(i)})^T u)^2
 \end{aligned}$$

Let $\mathcal{L} = [\sum_{i=1}^m ((x^{(i)})^T u)^2] - \lambda(u^T u - 1)$. Then

$$\frac{\partial \mathcal{L}}{\partial u} = \left[\sum_{i=1}^m 2x^{(i)}(x^{(i)})^T u \right] - 2\lambda u$$

Let $\frac{\partial \mathcal{L}}{\partial u} = 0$. Then

$$\begin{aligned}
 \left[\sum_{i=1}^m 2x^{(i)}(x^{(i)})^T u \right] - 2\lambda u &= 0 \\
 \sum_{i=1}^m x^{(i)}(x^{(i)})^T u &= \lambda u
 \end{aligned}$$

Note that

$$\begin{aligned}
 \Sigma &= \mathbb{E}[(x - \mu)(x - \mu)^T] \\
 &= \frac{1}{n} \sum_{i=1}^n x^{(i)}(x^{(i)})^T, \quad (\mu = 0)
 \end{aligned}$$

Therefore, as $\sum_{i=1}^m x^{(i)}(x^{(i)})^T u = \lambda u$

$$n\Sigma u = \lambda u$$

$$\Sigma u = \frac{\lambda}{n} u$$

That is to say, u is the first eigenvector of Σ , and therefore is the first principal component.

Problem 4

(a)

$$\begin{aligned}
\sum_{j=1}^d \log g'(w_j^T x^{(i)}) &= \sum_{j=1}^d \log \left(\frac{1}{\sqrt{2\pi}} \exp \left(-\frac{1}{2} (w_j^T x^{(i)})^2 \right) \right) \\
&= \sum_{j=1}^d -\frac{1}{2} (w_j^T x^{(i)})^2 - \frac{1}{2} \log(2\pi) \\
&= -\frac{1}{2} \|Wx^{(i)}\|_2^2 - \frac{d}{2} \log(2\pi) \\
\frac{\partial \ell}{\partial W} &= \sum_{i=1}^n W^{-T} - Wx^{(i)}(x^{(i)})^T \\
&= nW^{-T} - WX^T X
\end{aligned}$$

Let $\frac{\partial \ell}{\partial W} = 0$.

$$\begin{aligned}
nW^{-T} &= WX^T X \\
W^T W X^T X &= nI
\end{aligned}$$

Given any arbitrary orthogonal matrix $R \in \mathbb{R}^{d \times d}$ such that $R^T = R^{-1}$, then we have

$$W^T R^T R W X^T X = W^T W X^T X = I,$$

i.e., whenever we have solution W , there is always another solution RW . This is because since Gaussian distribution is rotational invariance, if applying a rotational transformation R on s , it is still Gaussian.

(b)

$$\begin{aligned}
\sum_{j=1}^d \log g'(w_j^T x^{(i)}) &= \sum_{j=1}^d -|w_j^T x^{(i)}| - \log 2 \\
&= -\|Wx^{(i)}\|_1 - d \log 2
\end{aligned}$$

As $\frac{\partial}{\partial w_j} \sum_{j=1}^d |w_j^T x^{(i)}| = \text{sign}(w_j^T x^{(i)}) \cdot x^{(i)}$,

$$\begin{aligned}
\frac{\partial}{\partial W} \sum_{j=1}^d \log g'(w_j^T x^{(i)}) &= \frac{\partial}{\partial W} \|Wx^{(i)}\|_1 \\
&= \begin{bmatrix} \text{sign}(w_1^T x^{(i)})(x^{(i)})^T \\ \text{sign}(w_2^T x^{(i)})(x^{(i)})^T \\ \vdots \\ \text{sign}(w_d^T x^{(i)})(x^{(i)})^T \end{bmatrix} \\
&= \text{sign}(Wx^{(i)})x^{(i)}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \ell}{\partial W} &= \sum_{i=1}^n W^{-T} - \frac{\partial}{\partial W} \|Wx^{(i)}\|_1 \\
&= \sum_{i=1}^n W^{-T} - \text{sign}(Wx^{(i)})x^{(i)}
\end{aligned}$$

Hence,

$$W := W + \alpha \left(W^{-T} - \text{sign}(Wx^{(i)})x^{(i)} \right)$$

(c) See the code.

Problem 5

(a)

$$\begin{aligned} B(V_1) - B(V_2) &= R(s) + \gamma \max_{a \in A} \sum_{s' \in S} P_{sa}(s') V_1(s') - R(s) - \gamma \max_{a \in A} \sum_{s' \in S} P_{sa}(s') V_2(s') \\ &= \gamma \max_{a \in A} \sum_{s' \in S} P_{sa}(s') V_1(s') - \gamma \max_{a \in A} \sum_{s' \in S} P_{sa}(s') V_2(s') \end{aligned}$$

Let $a_1 = \arg \max_{a \in A} \sum_{s' \in S} P_{sa}(s') V_1(s')$

$$\begin{aligned} B(V_1) - B(V_2) &= \gamma \max_{a \in A} \sum_{s' \in S} P_{sa}(s') V_1(s') - \gamma \max_{a \in A} \sum_{s' \in S} P_{sa}(s') V_2(s') \\ &\leq \gamma \sum_{s' \in S} P_{sa_1}(s') V_1(s') - \gamma \sum_{s' \in S} P_{sa_1}(s') V_2(s') \\ &= \gamma \sum_{s' \in S} P_{sa_1}(s') (V_1(s') - V_2(s')) \\ &\leq \gamma \sum_{s' \in S} P_{sa_1}(s') |V_1(s') - V_2(s')| \\ &\leq \gamma \sum_{s' \in S} P_{sa_1}(s') \max |V_1 - V_2| \\ &= \gamma \|V_1 - V_2\|_\infty \end{aligned}$$

Without losing generality, let $B(V_1) \geq B(V_2)$, and therefore

$$B(V_1) - B(V_2) = |B(V_1) - B(V_2)|$$

Since $B(V_1)(s) - B(V_2)(s) \leq \gamma \|V_1 - V_2\|_\infty$ holds for any s ,

$$\max_{s \in S} |B(V_1) - B(V_2)| \leq \gamma \|V_1 - V_2\|_\infty$$

Hence,

$$\|B(V_1) - B(V_2)\|_\infty \leq \gamma \|V_1 - V_2\|_\infty$$

(b) Assume there are two fixed points $B(V_1) = V_1$ and $B(V_2) = V_2$ for B . Therefore,

$$\begin{aligned} \|B(V_1) - B(V_2)\|_\infty &\leq \gamma \|V_1 - V_2\|_\infty \\ \|V_1 - V_2\|_\infty &\leq \gamma \|V_1 - V_2\|_\infty \\ (\gamma - 1) \|V_1 - V_2\|_\infty &\geq 0 \end{aligned}$$

As $\gamma < 1$, $\|V_1 - V_2\|_\infty = 0$. That is to say, V_1 and V_2 are the same fixed point, and there is only one fixed point for B .

Problem 6

(a) Converge in 140 trials.

(b) See Figure 2

(c) From Figures 3 to 5 we have observations:

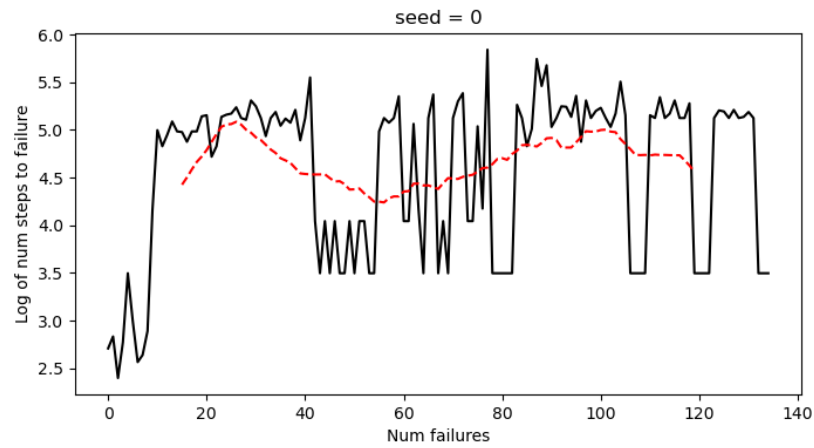


Figure 2: Learning curve with seed = 0

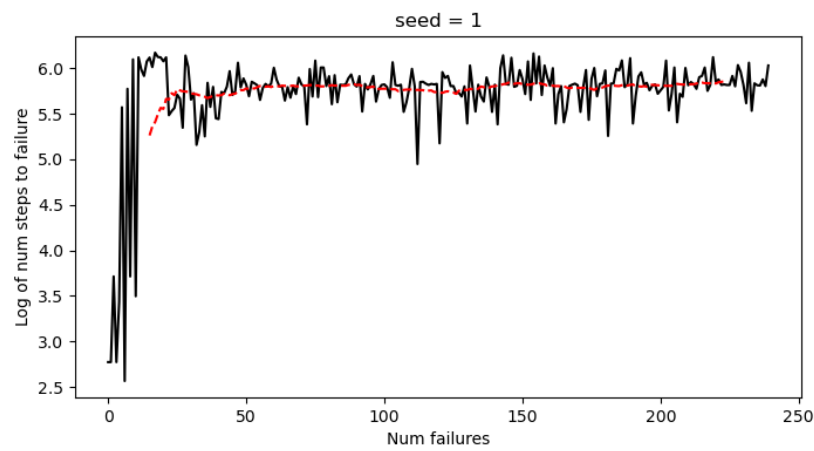


Figure 3: Learning curve with seed = 1

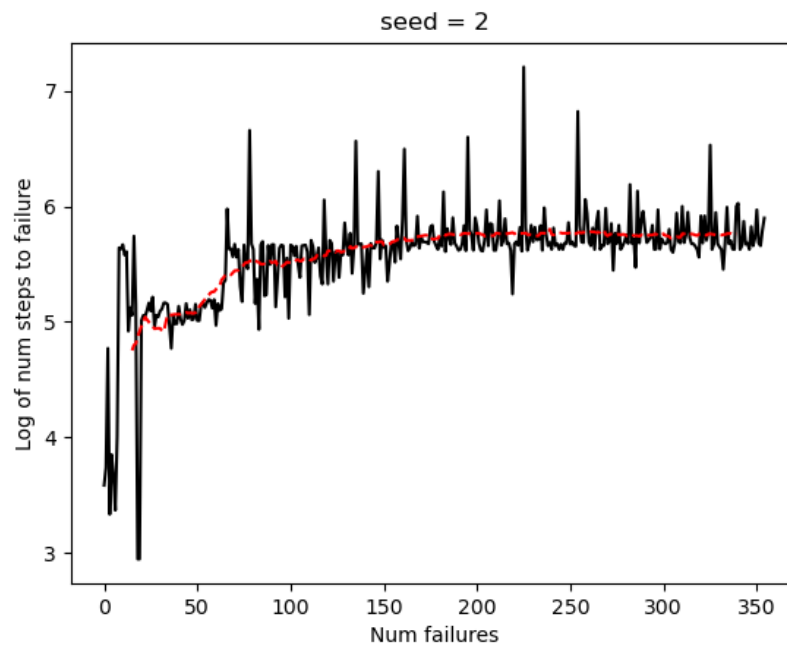


Figure 4: Learning curve with seed = 2

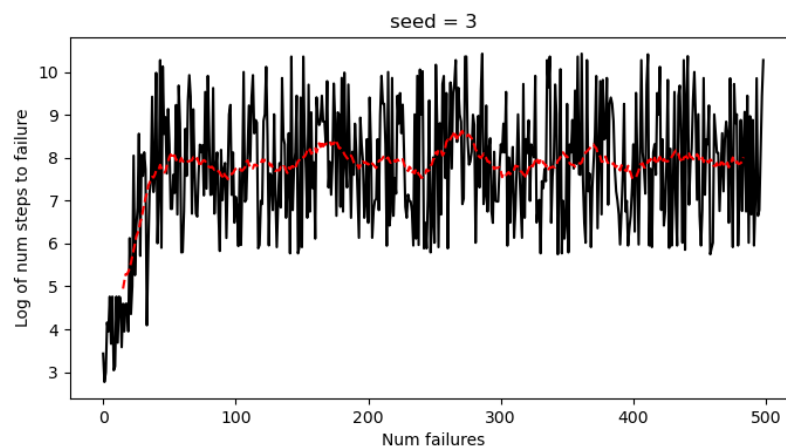


Figure 5: Learning curve with seed = 3

- Every seed shows a fast rise early on , where the algorithm quickly learns a working balancing policy.
- After that, every seed learns a stable policy where the red smoothed line flattens out and stays roughly constant.
- The exact numbers differ a lot between seeds: how many failures it takes to converge , how high the final policy is (log steps of -5.7 up to -8), and how noisy the black raw curve is during the flattened policy.

and What is implied about the algorithm:

- The algorithm reliably learns.
- The exact performance is random and depends on the choice of seed.